

No.of Functional Features Included in the Solution

Date	6 July 2026
TeamID	
ProjectName	Plugging into the Future : An Exploration of Electricity Consumption Patterns
MaximumMarks	5 Marks

FunctionalFeaturesOverview

This document lists all the functional features implemented in the CreditCardApproval Prediction System. Each feature contributes to providing an intelligent, reliable, and user-friendly web application for predicting credit card approval, storing prediction history, and displaying analytics through a dashboard.

S.No	Feature Name	Feature Description	Module / Component	Status (Done/In Progress /Pending)	Marks Contribution
1	Data Ingestion / Upload	Allows users to upload electricity consumption dataset (CSV files with hourly/daily usage).	HTML Form/Flask	Done	High
2	Data Preprocessing	Handles missing values, parses timestamps, and cleans electricity usage data.	Pandas/NumPy	Done	High
3	Feature Extraction	Extracts temporal features (hour of the day, day of the week, season) to analyze consumption patterns.	Pandas/Scikit-Learn	Done	High
4	Electricity Demand Forecasting	Predicts future electricity consumption using trained ML models (e.g., Random Forest/XGBoost/LSTM).	ML Model	Done	Very High
5	Consumption Analytics Dashboard	Displays interactive charts of usage trends, peak hours, and seasonal	Plotly / Matplotlib / Flask	Done	High

		variations.			
6	Responsive User Interface	Provides an intuitive and responsive web layout to visualize electricity insights.	HTML/CSS/ Bootstrap	Done	Medium
7	Model Loading	Loads the trained forecasting model (pkl/h5) for real-time predictions.	Pickle/Python	Done	High
8	Error Handling	Displays appropriate error messages for invalid file uploads or missing data columns.	Flask	Done	Medium
9	Model Accuracy / Metrics	Displays the accuracy or error metrics (like R-squared, RMSE) of the prediction model.	Scikit-learn / Flask	Done	High
10	Prediction History Logs	Stores and displays previous electricity usage and forecasting records.	SQLite + Flask	Done	High
11	Dashboard & Analytics	Displays total electricity consumption, peak demand hours, and seasonal usage analytics trends.	Flask + SQLite	Done	Very High

Feature Summary:

Metric	Count/ Value
Total Features Planned	11
Total Features Implemented	11
Core/Must-Have Features	7 (Input Form, Input Validation, Data Preprocessing, ML Prediction, Confidence Score, Result Display, Model Loading)

Additional/Nice-to-Have Features	4 Responsive and attractive web interface Dashboard with analytics and statistics Prediction history management SQLite database integration
Features Tested & Verified	Applicant input validation Machine learning model loading Electricity consumption prediction Prediction confidence score generation SQLite database storage Prediction history retrieval Dashboard analytics Result display Responsive user interface Application execution without runtime errors

Feature Category Breakdown:

S.No	Category	Featuresin Category	Example Features
1	User Inter face(UI)	3	Input Form, Result Page, Dashboard
2	Backend/ Logic	5	Input Validation, Data Preprocessing, ML Prediction , Confidence Score ,Error Handling
3	Database/ Storage	2	SQ Lite Database , Prediction History
4	API/Integration	0	Not Required
5	Security / Authentication	0	Not implemented (not required for this project)
6	Analytics/ Reporting	1	Dashboard Analytics